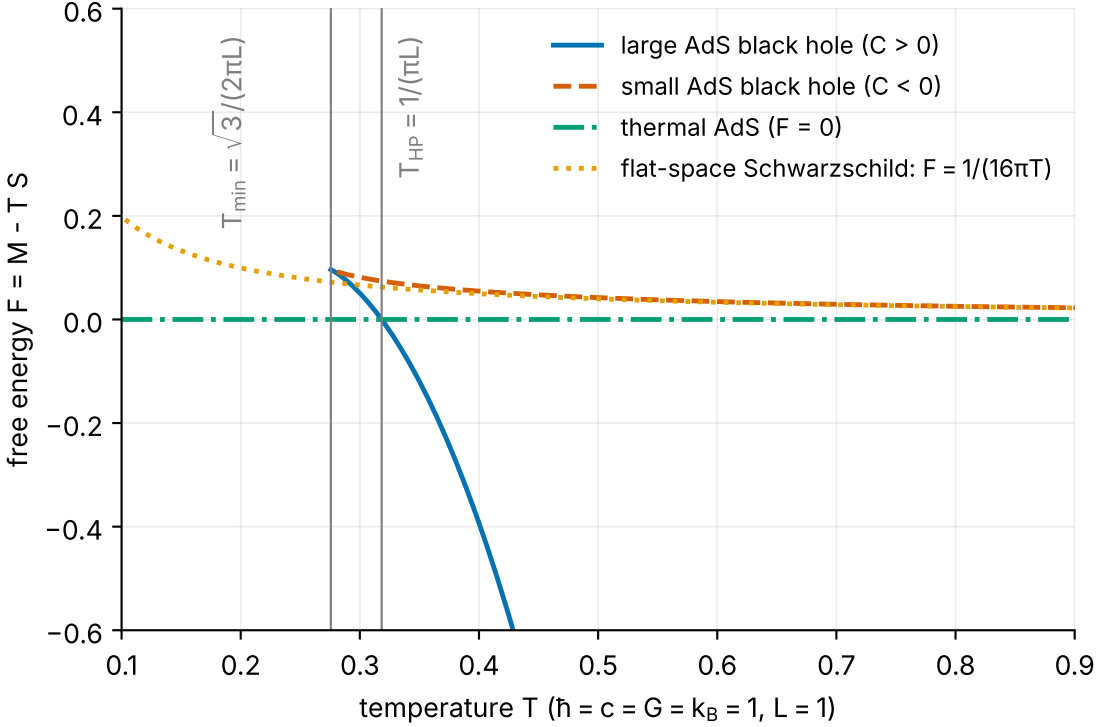
Hawking-Page transition in Schwarzschild-AdS$_4$
large AdS black hole (C > 0)
small AdS black hole (C < 0)
thermal AdS (F = 0)
flat-space Schwarzschild: F = 1/(16πT)
$T_{min} = \sqrt{3}/(2\pi L)$
$T_{HP} = 1/(\pi L)$
free energy F = M - T S
temperature T (ħ = c = G = k_B = 1, L = 1)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9